task_9.py - C:/Users/josth/AppData/Local/Programs/Python/Python313/task_9.py (3.13.17)

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# TASK 9: Implement and Test the Calculator

BASE_RATE = 30.00
FAT_STEP_VALUE = 0.40
SNF_STEP_VALUE = 0.30

REFERENCE_FAT = 3.5
REFERENCE_SNF = 8.5

MIN_FAT = 3.0
MAX_FAT = 6.0


def calculate_milk_rate(fat, snf):
    if not MIN_FAT <= fat <= MAX_FAT:
        raise ValueError("Fat value is outside the allowed range")

    if snf <= 0:
        raise ValueError("SNF must be positive")

    fat_steps = round((fat - REFERENCE_FAT) * 10)
    snf_steps = round((snf - REFERENCE_SNF) * 10)

    rate = (
        BASE_RATE
        + (fat_steps * FAT_STEP_VALUE)
        + (snf_steps * SNF_STEP_VALUE)
    )

    return round(rate, 2)


def calculate_payment(litres, fat, snf):
    if litres <= 0:
        raise ValueError("Milk quantity must be positive")

    rate = calculate_milk_rate(fat, snf)
    return round(litres * rate, 2)


def test_calculator():
    assert calculate_milk_rate(3.5, 8.5) == 30.00
    assert calculate_milk_rate(4.0, 8.7) == 32.60
    assert calculate_milk_rate(3.0, 8.0) == 26.50
    assert calculate_payment(10, 3.5, 8.5) == 300.00
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    assert calculate_payment(10, 3.5, 8.5) == 300.00

    try:
        calculate_payment(-5, 3.5, 8.5)
        assert False, "Negative quantity should be rejected"
    except ValueError:
        pass

    try:
        calculate_milk_rate(7.0, 8.5)
        assert False, "Out-of-range fat should be rejected"
    except ValueError:
        pass

    print("All tests passed successfully!")


# Test the calculator
test_calculator()

# Get user input
print("\n--- Milk Payment Calculator ---")

try:
    farmer_id = input("Enter Farmer ID: ")
    litres = float(input("Enter Milk Quantity in Litres: "))
    fat = float(input("Enter Fat Percentage: "))
    snf = float(input("Enter SNF Percentage: "))

    rate = calculate_milk_rate(fat, snf)
    payment = calculate_payment(litres, fat, snf)

    print("\nFarmer ID:", farmer_id)
    print("Milk Rate: Rs.", format(rate, ".2f"))
    print("Total Payment: Rs.", format(payment, ".2f"))

except ValueError as error:
    print("Invalid input:", error)
```